

Spain

Sovereign government data centre network — capacity, legal posture, provider landscape and migration path.

1 Starting point

Population	49.13 m
GDP	EUR 1,687 bn
Public administration (NACE O)	1,401 k
Electricity price	132.4 EUR/MWh
Renewables	59.7%
Live hyperscaler regions	3

2 What is structurally different

- Grid isolation. The national grid is an electrical island or nearly so (weak or single interconnection). The April 2025 Iberian blackout and Cyprus/Malta interconnector outages show the failure mode. Every site needs on-site generation and storage sized for multi-day ride-through, and the PUE and facility CAPEX assumptions should be revisited upward once site studies exist.
- Dense hyperscaler presence (3 live regions). Commercial capacity, fibre and skills exist in-country; the sovereign core can stay lean and the hybrid model works as designed. The risk is the opposite one: political pressure to declare a hyperscaler region 'sovereign enough' (Dutch GOAL.md section 17: location is not sovereignty).

3 Capacity

Servers	12,517 (CPU 8,731 / GPU 531 / storage 3,255)
IT critical load	20.1 MW
Facility design load	30.2 MW
Sites	4 (by capacity 3, floor 4)
Average per site	7.5 MW
Site count set by	the minimum-sites floor, not capacity
CAPEX	EUR 711 m
OPEX	EUR 61 m / yr

4 Proposed geography

Region	Role	Share	MW	Notes
Madrid	Primary civil cloud	36%	10.9	SGAD/AEAT/GISS estate, DE-CIX Madrid/ESpanix, two hyperscaler regions; connection queues.

Region	Role	Share	MW	Notes
Aragon / Zaragoza	Sovereign secondary	18%	5.4	AWS/Microsoft campuses prove grid; renewables, 300 km separation.
Catalonia / Barcelona	Government / continuity	18%	5.4	Mediterranean cable landings (2Africa, Medusa), CATNIX.
Andalusia / Seville - Malaga	Defense / industrial	18%	5.4	Southern separation, solar belt; water-stress cooling design.
Castilla y Leon - Galicia	Strategic reserve	10%	3.0	Wind/hydro surplus, Bilbao Atlantic landings nearby; reserve.

First-pass geographic hypotheses encoding only the obvious constraints, to be replaced by scored site selection.

5 Legal and regulatory posture

Governing instrument	Real Decreto 311/2022 (Esquema Nacional de Seguridad); Ley 40/2015 on the public sector
Cloud certification	ENS certification at Basic/Medium/High, with CCN-STIC guides; the most widely applied national scheme after SecNumCloud
Data classification	Ley de Secretos Oficiales: Difusion Limitada / Confidencial / Reservado / Secreto
Procurement route	Direccion General de Racionalizacion y Centralizacion de la Contratacion

Foreign jurisdiction exposure. AWS (Aragon), Azure and Google (Madrid) regions live; ENS High certification is the gate for government use

Under the US CLOUD Act and FISA 702 a provider subject to US jurisdiction can face a lawful order for data it holds regardless of where that data sits. Residency is necessary but not sufficient; what matters is who holds the keys and who can be compelled.

6 Current state and provider landscape

Government cloud	No single state cloud: Nube SARA / SGAD common services, sectoral ENS-Alta clouds (AEAT, GISS); regional sovereign clouds emerging (Madrid 2026); Telefonica/IBM, Indra, Oracle sovereign offers under ENS
Maturity	operational
Digital identity	CI@ve / DNle
Interconnection	DE-CIX Madrid and ESpanix; landings Bilbao (MAREA, Grace Hopper) and Barcelona (2Africa, Medusa)

7 Migration path and cost

Phase	Scope	MW	CAPEX	Cumulative	Hybrid
1	Sovereign core	8.0	EUR 153 m	22%	no
2	Security and defense	10.8	EUR 263 m	59%	no
3	State record	5.2	EUR 131 m	77%	no
4	Elective	6.2	EUR 164 m	100%	yes

Phase 1 is the number that matters: **EUR 153 m** for 8.0 MW, 22% of total CAPEX. That is the floor below which no hybrid arrangement helps — and it is a small fraction of the full build.

8 Geography and threat notes

Drought/heat and water stress (cooling constraint); low seismic except Granada-Murcia/Alboran; Iberian grid island (Apr 2025 nationwide blackout); long connection queues (Madrid, Aragon) despite cheap renewable power; abundant land; very low geopolitical risk